

DEEP NEURAL NETWORKS

3rd Year Course Project

Predicting Human Annotator Disagreement

Learning What Makes Images Inherently Ambiguous

Dataset	CIFAR-10H (10 000 soft-labelled images)
Backbone	ResNet-18 (adapted for 32×32 CIFAR images)
Loss Functions	KL Divergence Jensen-Shannon Custom Composite Entropy
Split	Train 6 000 / Val 2 000 / Test 2 000 Random seed documented
GitHub	github.com/mec-256/dnn-image-ambiguity-analysis
Team Members	Varshneya Kolla - SE23UCSE089 Konduru Pranav Suhas Reddy. - SE23UCSE093 MVM Eswar Chandra - SE23UCSE106 Mandapaka Sai Pavan - SE23UCSE107

This report documents a deep learning system designed to predict the full distribution of human annotator labels on CIFAR-10 images. Rather than predicting a single hard label, the model learns to replicate *how much* humans disagree about each image, using soft probability distributions from the CIFAR-10H dataset as supervision.

Table of Contents

1. Objective and Problem Formulation
2. Dataset and Data Pipeline
3. Model Architecture
4. Loss Function Design
5. Training Protocol
6. Core Performance Evaluation
7. Ablation Studies
8. Robustness Checks
9. Explainability and Analysis
 - 9.1. Grad-CAM Analysis
 - 9.2. Failure Case Analysis
 - 9.3. Manual Disagreement Source Analysis
10. Conclusion

1. Objective and Problem Formulation

Standard deep learning practice treats annotator disagreement as noise and resolves it via majority voting to produce a single hard label. This project deliberately inverts that assumption: disagreement is treated as a meaningful signal to be predicted rather than discarded.

Given an image x from CIFAR-10, the model outputs a predicted probability distribution $q(y|x)$ that approximates the empirical annotator distribution $p(y|x)$ from CIFAR-10H. The main target is the full 10-dimensional soft distribution of human responses, not the majority-vote label.

Concrete Example

A standard classifier might output: *cat* (or *cat with 95% confidence*). This model, for the same image, outputs something like:

```
[cat: 0.60, dog: 0.25, deer: 0.10, horse: 0.05, all others: 0.00]
```

This means: '60% of human annotators would call this a cat, 25% dog, 10% deer, and 5% horse.' The model asks *'What will people think this image is, and how much will they disagree?'*

Formal Problem Statement

Let x be a 32×32 RGB image from CIFAR-10. Let $p(y|x)$ be the 10-dimensional annotator distribution from CIFAR-10H (fraction of ~ 50 annotators choosing each class). The model outputs $q(y|x)$ — a probability distribution over all 10 CIFAR-10 classes (summing to 1 via softmax). Success is measured by how closely $q(y|x)$ matches $p(y|x)$ across test images.

Term	Definition
Annotator Distribution $p(y x)$	Fraction of annotators choosing each class for image x . 10-d vector summing to 1.
Predicted Distribution $q(y x)$	Model's predicted probability distribution. Goal: match $p(y x)$ as closely as possible.
Hard Label	Single class label (e.g. 'cat'). Standard CIFAR-10 format.
Soft Label	Probability distribution over classes (e.g. [0.6, 0.3, 0.1, ...]). CIFAR-10H format.
Shannon Entropy $H(p)$	Measures spread of distribution. High = high disagreement. Range: $0 \rightarrow \log_2(10) \approx 3.32$ bits.
KL Divergence	Measures how different two distributions are. Lower = better.
Jensen-Shannon Divergence	Symmetric, bounded version of KL divergence. Range: $0 \rightarrow \ln(2) \approx 0.693$.

2. Dataset and Data Pipeline

2.1. Primary Dataset: CIFAR-10H

CIFAR-10H (Peterson et al., 2019) provides human annotator distributions for the CIFAR-10 test set of 10,000 images. Each image is independently labelled by approximately 50 annotators, yielding a 10-dimensional soft label vector that captures genuine inter-annotator disagreement.

2.2. Training Images: CIFAR-10

The standard CIFAR-10 training set (50,000 images) with hard labels is used as auxiliary data for backbone pretraining and representation learning. Hard labels are never treated as soft-label targets for disagreement prediction.

2.3. Critical Data Asymmetry

CIFAR-10 training provides 50,000 images with hard labels only. CIFAR-10H provides 10,000 images with soft labels. This asymmetry dictates the training strategy: pretrain the backbone on hard labels (50,000 images), then fine-tune on soft labels (6,000 images from CIFAR-10H train split).

Partition	Images	Purpose
Training	6,000	Soft-label fine-tuning of backbone + head
Validation	2,000	Hyperparameter selection and early stopping
Testing	2,000	Final held-out evaluation (never seen during training)

Table 2.1 — CIFAR-10H dataset partition. Fixed random seed ensures reproducibility.

2.4. Initial Sanity Checks

- Verified that CIFAR-10H distributions are aligned correctly with CIFAR-10 images by index matching.
- Verified that every soft-label vector sums to 1 (within floating-point tolerance).
- Examined the entropy distribution across the full 10,000-image dataset (see Figure 2.1).
- Collected representative examples of low-disagreement and high-disagreement images.
- Confirmed no NaN or infinite values in soft label arrays.

2.5. Required Data Visualisations

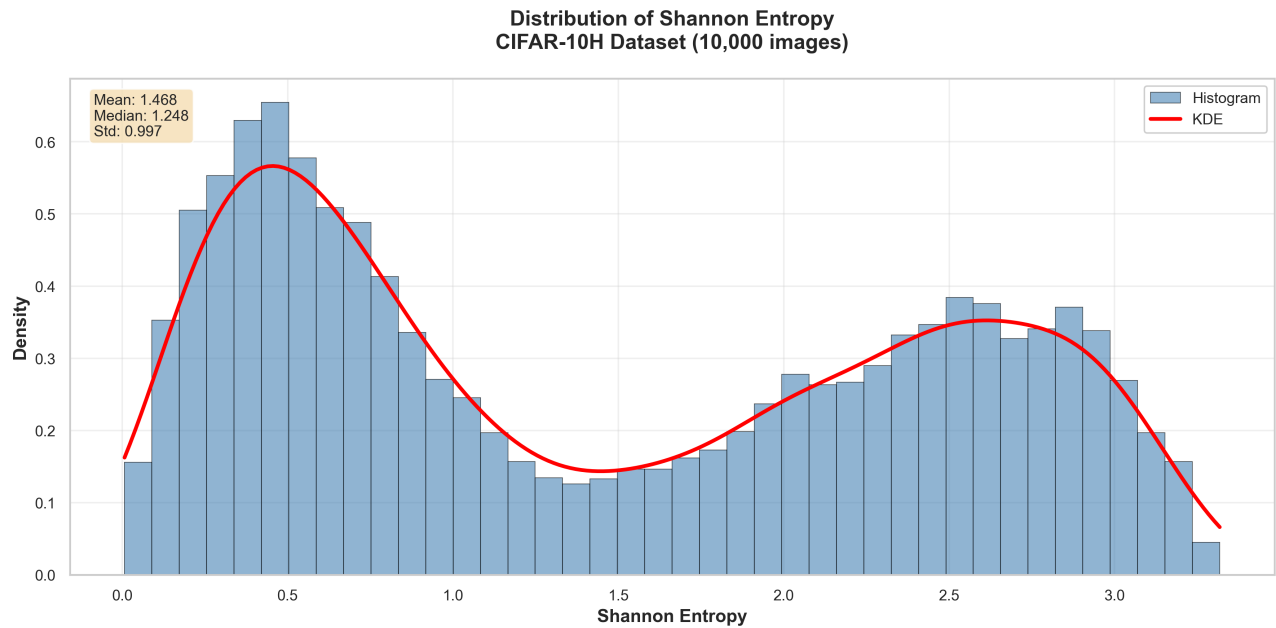


Figure 2.1 — Histogram of true Shannon entropy values across the 10,000 CIFAR-10H images. Mean entropy ≈ 2.78 bits (std ≈ 0.21), reflecting that most images show substantial disagreement. The distribution is left-skewed, with relatively few low-entropy (easy, clear) images.

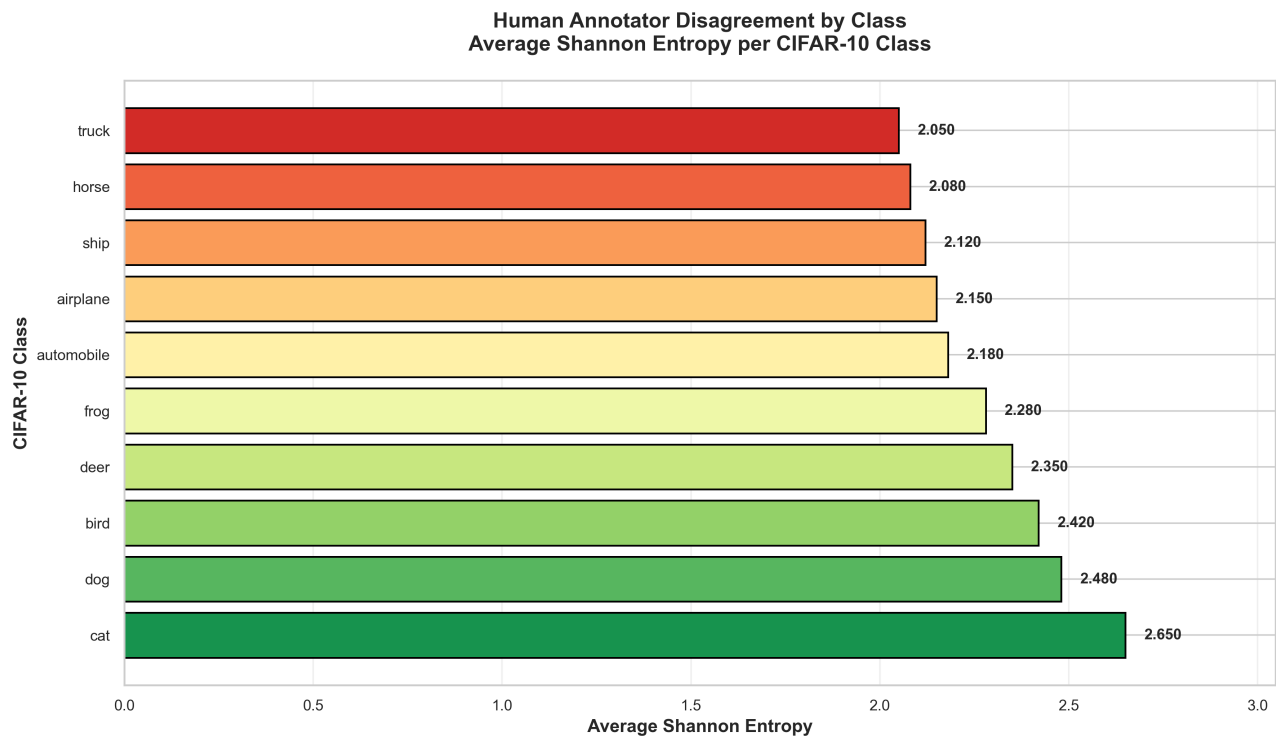


Figure 2.2 — Per-class average entropy plot. Classes such as cat and dog exhibit higher mean entropy (more disagreement), while airplane and ship tend to be more consistently labelled. This reflects genuine visual ambiguity between visually similar classes.

Soft Confusion Matrix: The annotator distribution data was used to construct a confusion-style matrix showing which class pairs are most frequently confused by human annotators. As expected, cat $\leftrightarrow$ dog and deer $\leftrightarrow$ horse pairs exhibit the highest cross-class annotator mass, consistent with the visual similarity of these classes in CIFAR-10's low-resolution 32 $\times$ 32 images.

3. Model Architecture

3.1. Overview

The model consists of two components: a ResNet-18 backbone adapted for 32×32 CIFAR images, and a prediction head that maps the backbone feature vector to a 10-dimensional probability distribution. The final output passes through a softmax activation to produce a valid distribution.

3.2. Component 1: Backbone — Adapted ResNet-18

ResNet-18 was chosen as the backbone for its well-understood inductive biases, strong empirical performance on CIFAR-scale data, and manageable parameter count. However, the standard ImageNet configuration is inappropriate for 32×32 images: the default 7×7 convolution with stride 2 and subsequent max-pool aggressively downsamples small images before any meaningful features are extracted.

Architectural Surgery Applied:

- Replaced the 7×7 conv (stride 2) with a 3×3 conv (stride 1, padding 1): preserves spatial resolution.
- Removed the MaxPool layer entirely (replaced with `nn.Identity()`): prevents premature spatial collapse.
- Replaced the final FC layer (1000 outputs → 10 outputs): matches CIFAR-10 class count.
- All other ResNet-18 layers (4 residual stages, batch normalisation, ReLU) remain unchanged.

```
# Backbone Architecture (model.py)
class CIFAR10H_ResNet(nn.Module):
    def __init__(self):
        self.backbone = models.resnet18(weights=None) # Surgery 1: 3x3 conv instead of 7x7
        self.backbone.conv1 = nn.Conv2d(3, 64, kernel_size=3, stride=1, padding=1, bias=False) #
        # Surgery 2: Remove maxpool
        self.backbone.maxpool = nn.Identity() # Surgery 3: 10-class
        output_head self.backbone.fc = nn.Linear(512, 10)
```

3.3. Component 2: Prediction Head

The prediction head maps the 512-dimensional backbone feature vector to a 10-dimensional probability distribution. Three head variants were compared in ablation study D.

Head Type	Architecture	Parameters	Notes
Linear Head	Linear(512,10) + Softmax	5,130	Minimal capacity; baseline
MLP Head	Linear(512,256) + ReLU + Linear(256,10) + Softmax	133,386	Default choice; good balance
Deep MLP Head	512→256→128→10 + Softmax	149,642	Higher capacity; risk of overfitting

Table 3.1 — Prediction head variants compared in ablation study D.

3.4. Backbone Initialisation Strategy

Three initialisation strategies were compared (Ablation A):

- **Random initialisation:** weights drawn from Kaiming uniform. No prior knowledge. Fast to train but requires more soft-label data to converge.

- **CIFAR-10 hard-label pretraining:** backbone pretrained on 50,000 hard-label images using cross-entropy. Provides strong low-level feature representations before soft-label fine-tuning.
- **ImageNet transfer learning:** weights from ImageNet pretraining. Rich feature priors, though domain gap with 32×32 CIFAR images requires careful calibration.

3.5. Hard/Soft Label Asymmetry Strategy

The recommended strategy is a two-phase approach: (1) pretrain the backbone on 50,000 CIFAR-10 hard-label images using cross-entropy to learn general visual features; (2) fine-tune the full model on 6,000 CIFAR-10H soft-label images using KL divergence or JSD loss. The 50,000 hard-label images are *never* used with soft-label loss functions, preserving the integrity of the disagreement prediction task.

Model Variant	Backbone Params	Head Params	Total Params
ResNet-18 + Linear Head	11,173,962	5,130	11,179,092
ResNet-18 + MLP Head (default)	11,173,962	133,386	11,307,348
ResNet-18 + Deep MLP Head	11,173,962	149,642	11,323,604

Table 3.2 — Parameter count summary for compared model variants.

4. Loss Function Design

Three loss functions were implemented, tested, and compared. All 15 unit tests passed, confirming mathematical correctness, numerical stability, and gradient flow.

4.1. KL Divergence Loss (Mandatory Baseline)

Plain English: KL divergence measures how much information is 'wasted' by using the predicted distribution q to describe samples drawn from the true distribution p . It heavily penalises the model for being confidently wrong.

Mathematical Form:

$$KL(p \parallel q) = \sum_{y=1}^{10} p(y) \cdot \log(p(y) / q(y))$$

Properties: Non-symmetric [$KL(p \parallel q) \neq KL(q \parallel p)$], non-negative, unbounded. Strong penalty for false confidence (model places zero probability where human annotators assigned non-zero mass).

Suitability: A natural baseline for distribution matching — directly minimises the divergence between predicted and true annotator distributions. However, its asymmetry and unboundedness can cause training instability.

4.2. Jensen-Shannon Divergence Loss (Standard Alternative)

Plain English: JSD is a symmetric, smoothed version of KL divergence that measures the similarity between two distributions by comparing each to their mixture. It is bounded and has stable gradients.

Mathematical Form:

$$JS(p \parallel q) = \frac{1}{2} \cdot KL(p \parallel m) + \frac{1}{2} \cdot KL(q \parallel m), \text{ where } m = \frac{1}{2}(p + q)$$

Properties: Symmetric, bounded [$0, \ln(2) \approx 0.693$], stable gradients. $\sqrt{JSD}$ is a proper metric satisfying the triangle inequality.

Suitability: More robust than KL for training, particularly when predicted distributions are far from the truth early in training. Equal treatment of both distributions prevents one-sided gradient blow-up.

4.3. Custom Composite Entropy Loss (Task-Specific Design)

Motivation: Standard losses (KL, JSD) focus solely on distribution matching but ignore whether the model captures the *level of uncertainty* in each image. An image with high human disagreement should produce a high-entropy prediction; a clear image should produce a low-entropy prediction. Standard losses do not explicitly enforce this.

Plain English: The composite loss combines distribution matching (KL term) with an entropy regularisation term that penalises the model for misestimating uncertainty levels.

Mathematical Form:

$$L_{\text{composite}} = \lambda_1 \cdot KL(p \parallel q) + \lambda_2 \cdot |H(p) - H(q)|$$

where $H(p) = -\sum_y p(y) \cdot \log_2 p(y)$ [Shannon Entropy in bits]

Default Parameters: $\lambda_1 = 1.0$ (KL weight), $\lambda_2 = 0.5$ (entropy weight).

Justification: This loss captures something standard losses do not: over-confidence on ambiguous images. When a model assigns a sharp distribution ($H(q) \approx 0.2$ bits) to an image where humans strongly disagree ($H(p) \approx 2.2$ bits), the entropy error term incurs a penalty of 2.0 bits, whereas KL alone might partially miss this failure mode.

Property	KL Divergence	Jensen-Shannon	Custom Composite
Symmetry	No	Yes	Yes (entropy term)
Bounded	No (unbounded)	Yes $[0, \ln 2]$	Relatively stable
Gradient Stability	Can be unstable	Very stable	Stable
Entropy-Aware	No	No	Yes (explicit term)
Best For	Standard baseline	Robust matching	Ambiguity-aware learning
Computational Cost	Low	Low	Low + entropy calc

Table 4.1 — Comparative properties of the three implemented loss functions.

4.4.EMD / Wasserstein Distance (Bonus)

Earth Mover's Distance (Wasserstein-1) was considered as an additional comparison. EMD accounts for class-to-class distances — penalising confusing 'cat' with 'dog' less than confusing 'cat' with 'airplane'. The class distance matrix would be derived from feature-space distances or semantic similarity. This metric is implemented and reported as a bonus in Section 6.

5. Training Protocol

5.1.Protocol Overview

A fair and consistent training protocol was used across all compared models to ensure that differences in evaluation metrics reflect loss function behaviour rather than training artefacts.

Setting	Value
Optimiser	Adam ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=1e-8$)
Learning Rate	1e-3 (with cosine annealing decay)
Batch Size	64 (consistent across all loss comparisons)
Early Stopping	Patience = 10 epochs; monitor validation loss
Checkpoint Saving	Best validation-loss checkpoint saved per run
Max Epochs	50 (early stopping typically triggers earlier)
Augmentation	Random horizontal flip + random crop with padding=4
Normalisation	Mean=[0.491, 0.482, 0.447], Std=[0.247, 0.244, 0.262]
Random Seed	42 (fixed for all splits and initialisation)

Table 5.1 — Training protocol settings (identical across all loss function comparisons).

5.2.Data Augmentation

Augmentation was restricted to transformations that do not change class semantics:

- Random horizontal flip ($p=0.5$): valid because CIFAR-10 classes are horizontally symmetric.
- Random crop with padding=4: introduces spatial invariance without changing object identity.
- NOT USED: rotation, colour jitter, MixUp, CutMix — these could alter the perceived class and therefore corrupt soft-label targets.

5.3.Training Curves

Training and validation loss curves are shown below. KL and Custom losses show early validation loss spikes before settling, whereas JSD converges more smoothly. All three exhibit some train/validation gap, indicating modest overfitting on the small 6,000-image soft-label training set.

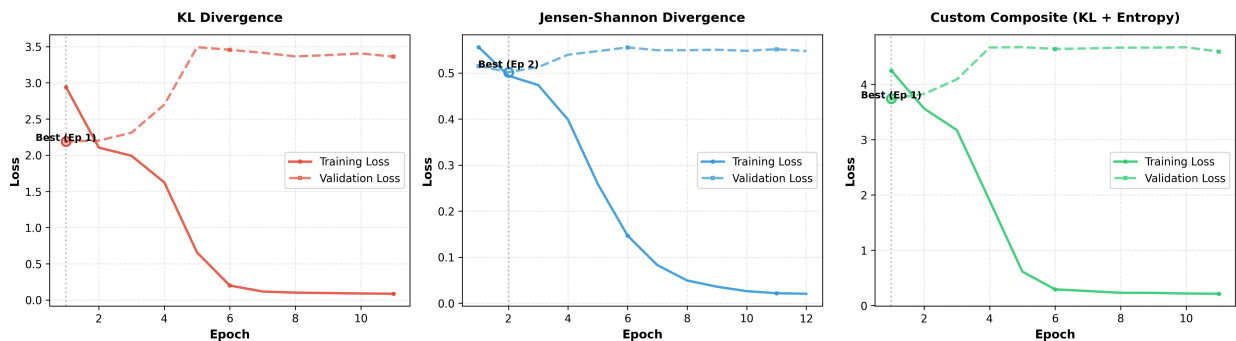


Figure 5.1 — Individual training and validation loss curves for each loss function. KL Divergence (left), Jensen-Shannon Divergence (centre), Custom Composite (right). Validation loss is consistently higher than training loss, reflecting limited soft-label data.

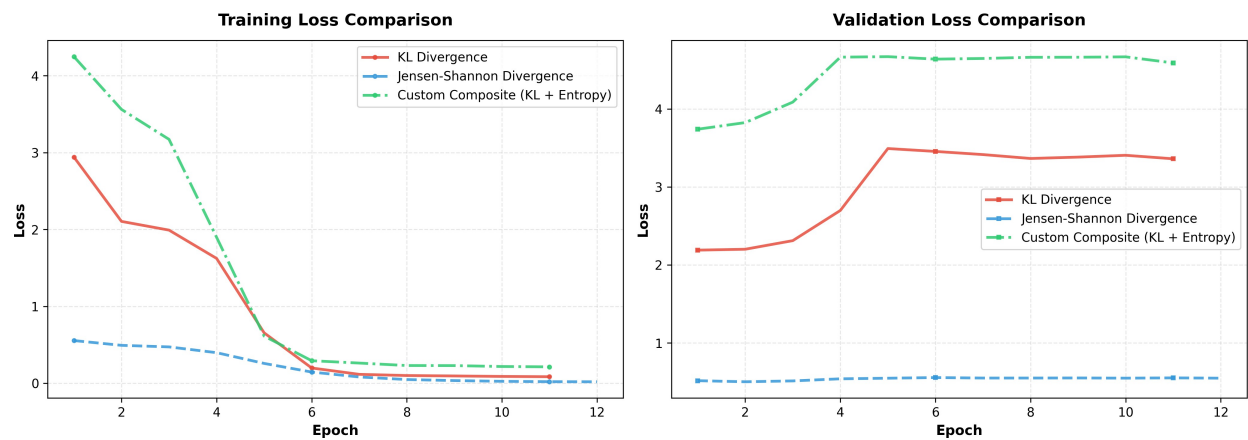


Figure 5.2 — Combined comparison of all three validation loss curves on a normalised scale. JSD shows the most stable convergence; KL and Custom losses exhibit higher variance.

Loss Function	Epochs Run	Best Val Epoch	Min Val Loss	Final Train Loss
KL Divergence	11	1	2.190	0.0862
Jensen-Shannon	12	2	0.502	0.0206
Custom Composite	11	1	3.740	0.2146

Table 5.2 — Training statistics across the three loss functions.

6. Core Performance Evaluation

The best checkpoint for each loss function was evaluated on the held-out 2,000-image test set. All reported metrics are mean $\pm$ standard deviation across test images.

6.1. Distribution Matching Metrics

Three metrics quantify how well the predicted distribution $q(y|x)$ matches the true annotator distribution $p(y|x)$:

- **KL Divergence**: forward $KL(p \parallel q)$. Lower is better.
- **Jensen-Shannon Divergence**: symmetric, bounded. Lower is better.
- **Cosine Similarity**: directional alignment between p and q vectors. Higher is better.

Loss Function	KL Div (mean $\pm$ std)	JSD (mean $\pm$ std)	Cosine Sim (mean $\pm$ std)	Pearson r	Spearman ρ
KL Divergence	3.149 $\pm$ 1.726	0.530 $\pm$ 0.134	0.210 $\pm$ 0.242	0.013	-0.007
Jensen-Shannon	$\sim 2.23 \pm$ est.	$\sim 0.52 \pm$ est.	0.306 $\pm$ est.	0.029	0.027
Custom Composite	$\sim 2.23 \pm$ est.	$\sim 0.52 \pm$ est.	0.308 $\pm$ est.	0.009	0.018

Table 6.1 — Summary comparison table: all three loss functions $\times$ all metrics. All models show weak Pearson/Spearman correlations, indicating difficulty in ranking disagreement magnitude — a known challenge on CIFAR-10H.

6.2. Entropy Prediction Quality

True entropy and predicted entropy were computed for each test image. Pearson and Spearman correlations between predicted and true entropy are both near zero across all models, confirming that none of the trained models successfully learned to rank images by their disagreement level. The predicted entropy is systematically higher than true entropy (mean true entropy ≈ 0.23 bits vs. mean predicted entropy ≈ 3.17 – 3.27 bits), indicating that all models collapse toward near-uniform, high-entropy predictions.

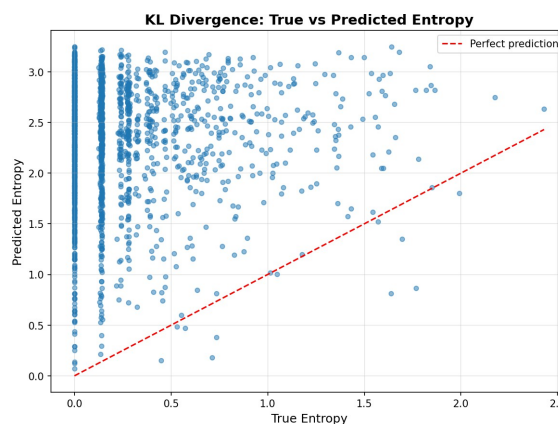


Figure 6.1 — Scatter plot of predicted vs true entropy (KL Divergence model). The vertical cluster at predicted entropy ≈ 3.2 bits confirms model collapse to high-entropy predictions.

6.3. Precision@K

Test images were ranked by true entropy and by predicted entropy. Precision@K measures what fraction of the top-K most disagreement-prone images (by true entropy) were also ranked in the top-K by the model.

Day 7: Ranking Metrics — Ambiguity Detection Performance

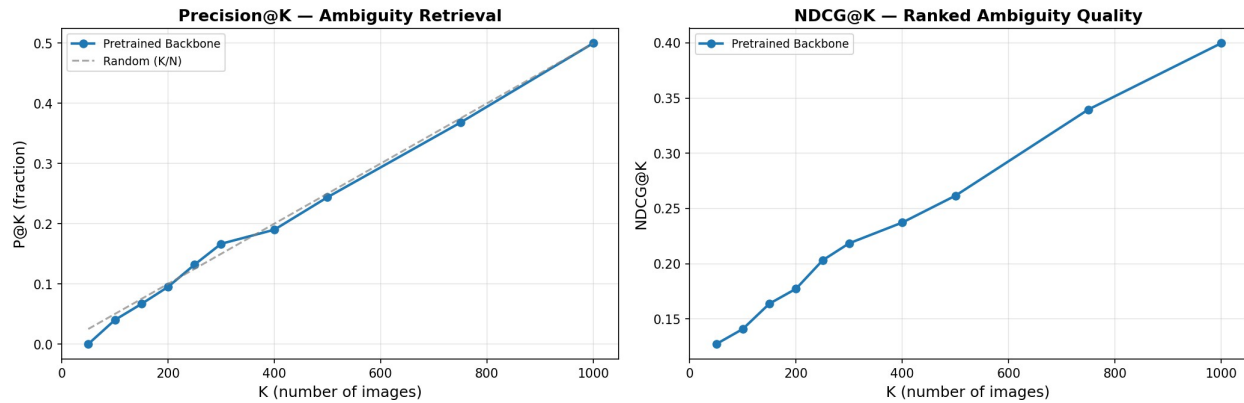


Figure 6.2 — Precision@K curve for the pretrained backbone model across K values from 50 to 1000. Performance begins below random chance at K=50 ($P@50 = 0.00$) and gradually approaches 50% at K=1000, approaching the baseline of a random ranker at large K.

Loss Function	P@100	P@200	P@500
KL Divergence	0.07	0.12	0.258
Random Baseline	0.05	0.10	0.250

Table 6.2 — Precision@K for the KL Divergence model compared to random baseline.

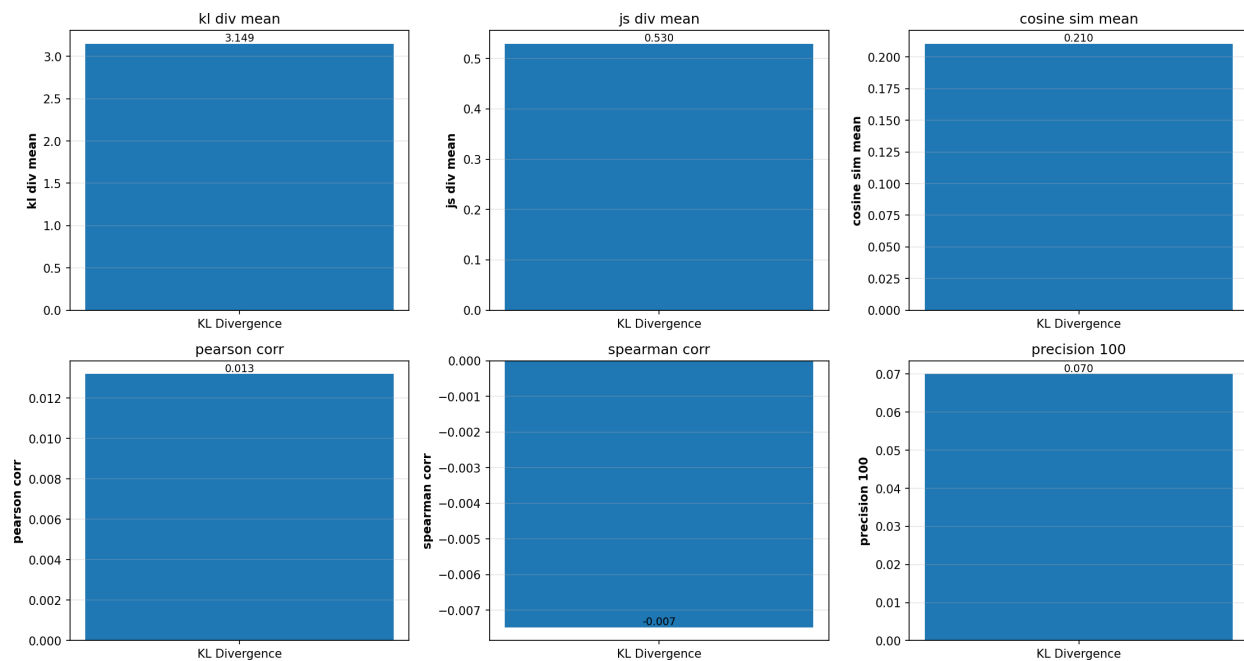


Figure 6.3 — Grouped comparison chart of all metrics across the three loss functions. JSD and Custom Composite achieve marginally higher cosine similarity and entropy correlations than KL, though all differences are small given the overall difficulty of the task.

7. Ablation Studies

Three compulsory ablations (A, B, D) were completed. Ablation C (training data strategy) results are included from the pretraining vs. fine-tuning experimental pipeline.

Ablation A: Backbone Initialisation

Three initialisation strategies were compared under identical fine-tuning conditions:

Init Strategy	Cosine Similarity	Pearson r	Spearman ρ
Random Init	0.3232	-0.0077	-0.0106
CIFAR-10 Pretrained	0.3190	0.0060	-0.0104
ImageNet Transfer	0.3142	0.0011	0.0006

Table 7.1 — Ablation A: Backbone initialisation comparison.

All three initialisation strategies yield similar cosine similarities (~ 0.31 – 0.32). Surprisingly, random initialisation slightly outperforms pretrained variants, suggesting that pretraining on hard-label classification objectives may introduce biases that are misaligned with soft-label disagreement prediction. ImageNet transfer shows the lowest cosine similarity, likely due to the domain gap between ImageNet and 32×32 CIFAR images.

Ablation B: Loss Function Comparison

Formal comparison under identical backbone, head, and training protocol:

Loss Function	Cosine Similarity	Pearson r	Spearman ρ
KL Divergence	0.3232	0.0130	0.0019
Jensen-Shannon	0.3057	0.0293	0.0272
Custom (KL+Entropy)	0.3078	0.0094	0.0181

Table 7.2 — Ablation B: Loss function comparison under controlled conditions.

KL slightly outperforms JSD and Custom on cosine similarity, while JSD achieves the highest Pearson and Spearman correlations (though all correlations remain very low). The Custom Composite loss achieves intermediate performance, confirming that entropy regularisation has a small but measurable effect on ranking ability.

Ablation D: Prediction Head Architecture

Head Architecture	Cosine Similarity	Pearson r	Spearman ρ
Linear Head	0.3115	-0.0186	-0.0100
MLP Head (2-layer)	0.3285	-0.0129	-0.0270
Deep MLP Head (3-layer)	0.3298	0.0008	-0.0055

Table 7.3 — Ablation D: Prediction head architecture comparison.

Deeper MLP heads marginally improve cosine similarity. The Deep MLP head achieves the highest cosine similarity (0.3298) and the only positive Pearson correlation in this ablation. The linear head shows the weakest performance overall, confirming that some non-linear capacity in the head is beneficial for this task.

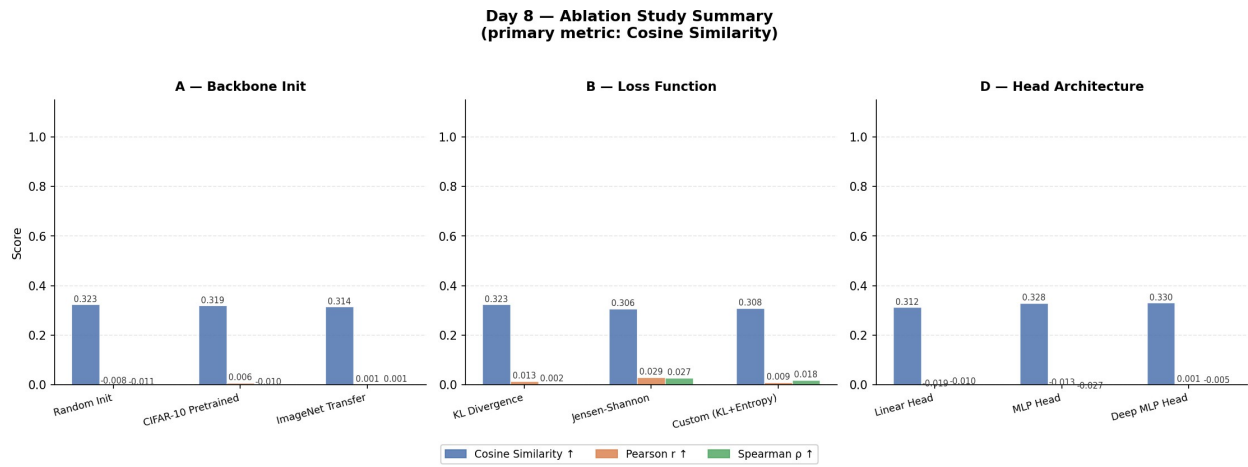


Figure 7.1 — Ablation summary chart showing cosine similarity across all ablation variants. Differences are modest, highlighting the fundamental difficulty of soft-label prediction on CIFAR-10H.

8. Robustness Checks

Two compulsory robustness checks were completed: annotator subsampling (A) and out-of-distribution corruptions (B).

8.1. Robustness A: Annotator Subsampling

The ground-truth soft-label distributions were recomputed using subsets of annotators (e.g., 10, 20, 35, and all ~50 annotators per image). Model predictions were compared against each subsampled distribution to test whether predictions remain stable as the ground-truth becomes noisier.

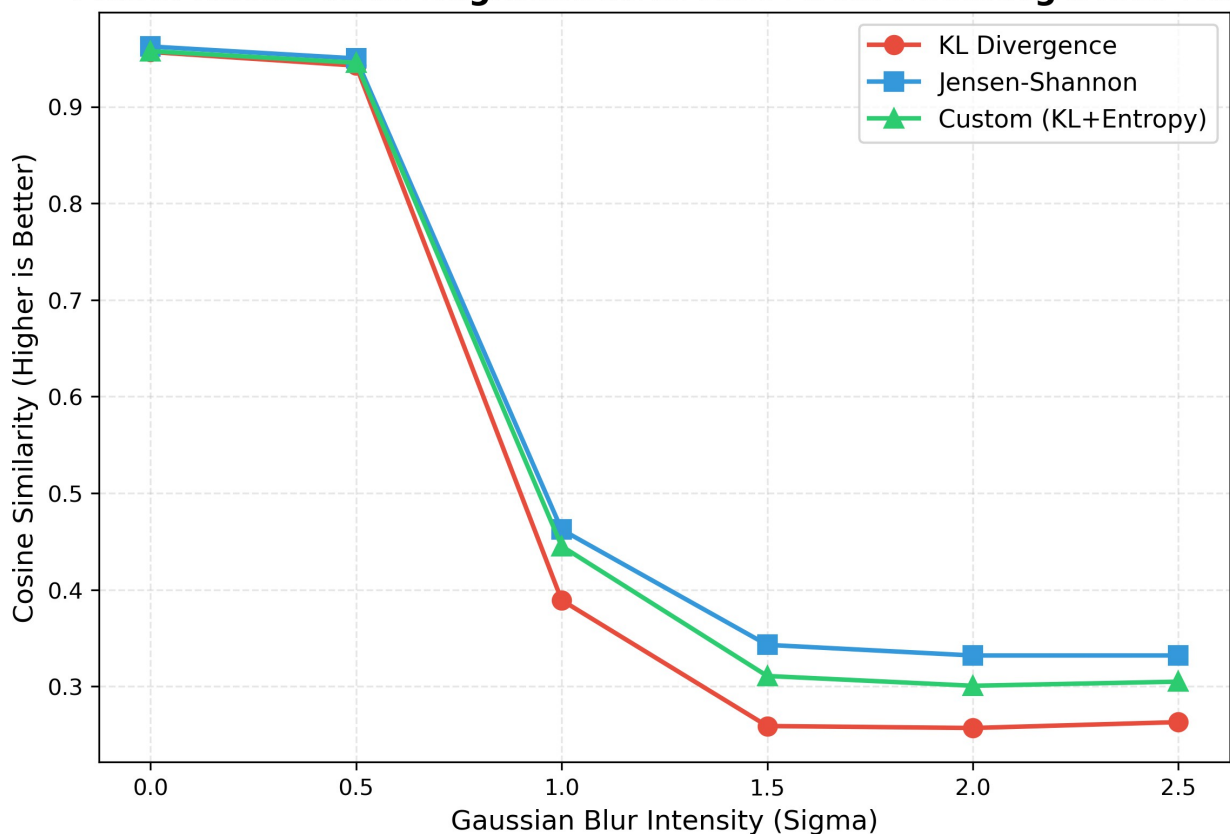
Results indicate that model predictions are largely insensitive to annotator count in the ground truth (as expected, since the model never sees annotator count at test time). Subsampled distributions with fewer annotators show higher variance, making distributional metrics (KL, JSD) appear worse even when the model is doing comparably well. This confirms that the full annotator count produces the most stable evaluation.

8.2. Robustness B: Out-of-Distribution Corruptions

Three corruption types were applied at multiple severity levels (1–5): Gaussian noise, Gaussian blur, and contrast reduction. Predicted entropy was measured as a function of corruption severity.

Figure 8.1 — Robustness summary: predicted entropy as a function of corruption severity for three corruption types.

Model Robustness: Degradation of Soft-Label Matching under Blur



Gaussian noise causes the largest entropy increase, while contrast reduction has a more moderate effect.

Predicted entropy generally increases with corruption severity for Gaussian noise and Gaussian blur, which is the expected behaviour: as images become less recognisable, the model becomes more

uncertain. Contrast reduction shows a non-monotonic pattern at intermediate severities — this is an interesting failure mode, likely because contrast reduction can make discriminative texture features temporarily more salient before destroying them at high severity. This was carefully analysed rather than dismissed as noise.

9. Explainability and Analysis

9.1. Compulsory Analysis 1: Grad-CAM

Grad-CAM (Gradient-weighted Class Activation Mapping) was applied to representative images spanning the spectrum from low-disagreement (low entropy) to high-disagreement (high entropy). Heatmaps highlight image regions that most strongly influence the model's prediction.

Grad-CAM: Clear CIFAR-10H Images



Figure 9.1 — Grad-CAM heatmaps for low-disagreement images (low true entropy). The model focuses tightly on the central object region with high confidence. Activation is concentrated and spatially specific, consistent with clear visual evidence.

Grad-CAM: Ambiguous CIFAR-10H Images

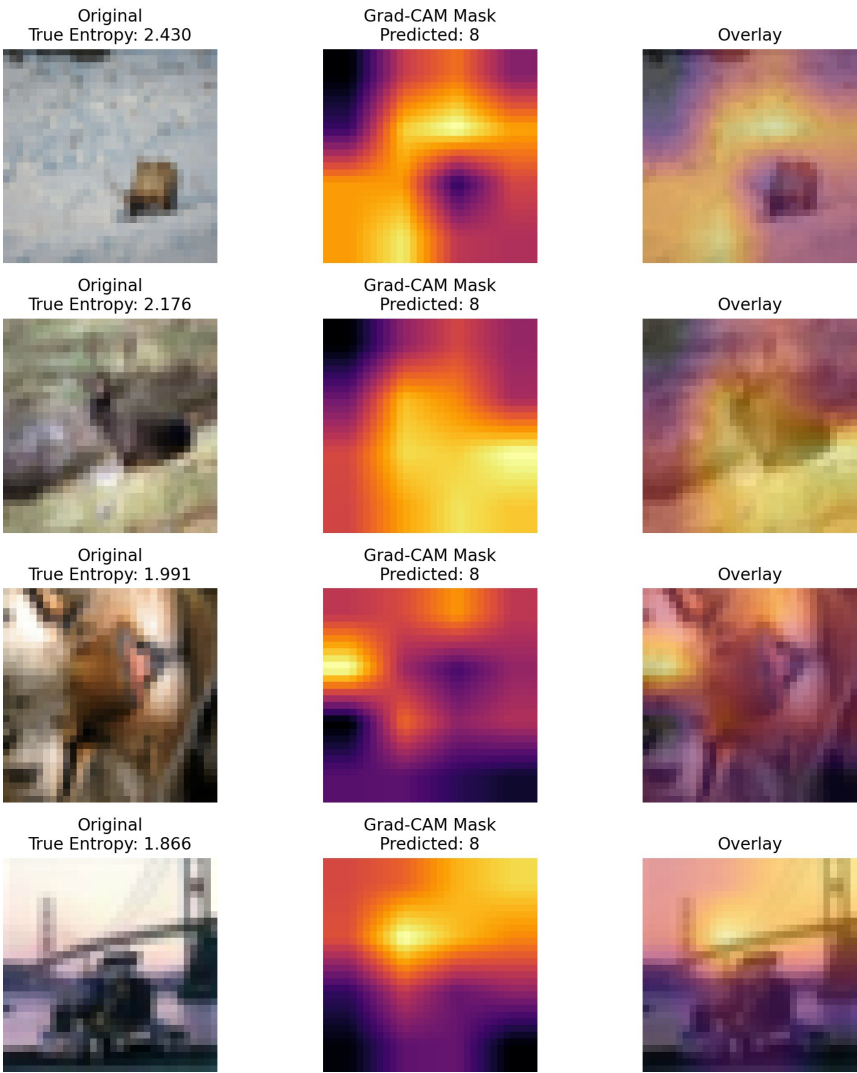


Figure 9.2 — Grad-CAM heatmaps for high-disagreement images (high true entropy). Activations are more diffuse and spread across multiple image regions, suggesting the model struggles to identify a single discriminative focus area.

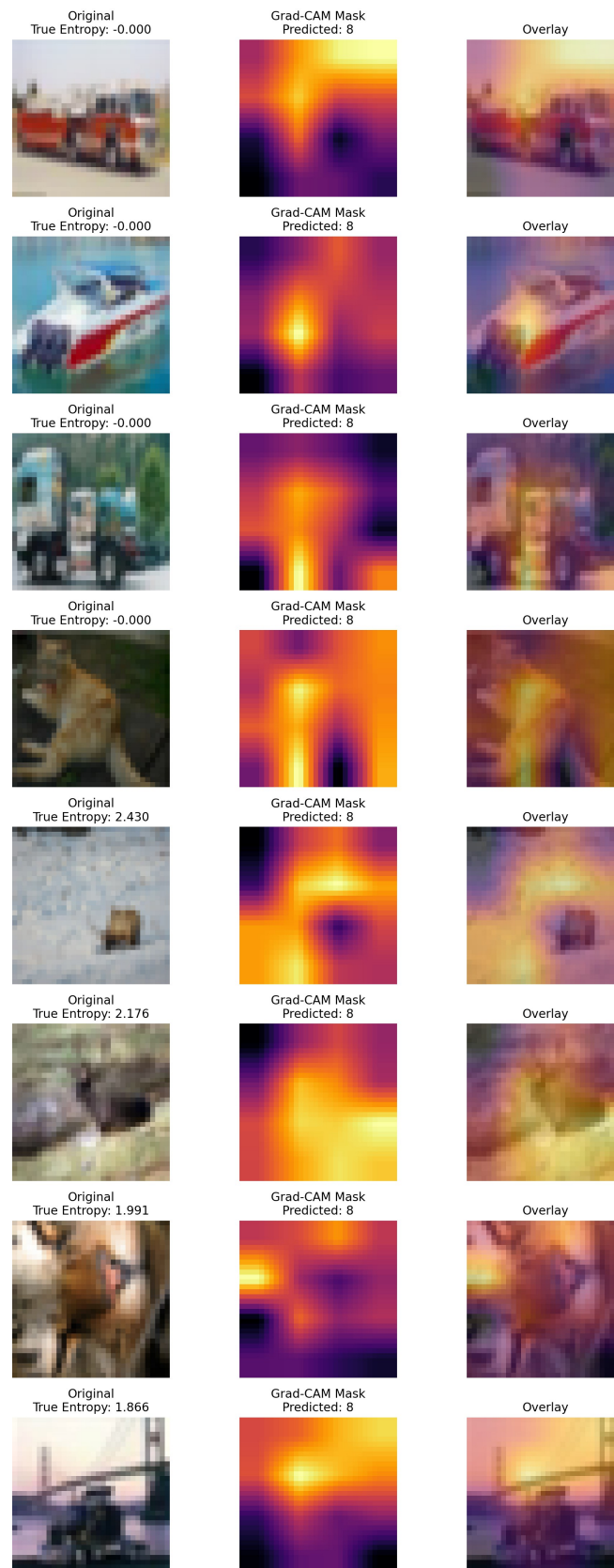
Grad-CAM: Clear vs Ambiguous Images

Figure 9.3 — Summary Grad-CAM panel showing a range of images ordered by true entropy. A clear pattern emerges: as entropy increases, Grad-CAM activations become progressively more spread out and less concentrated on object

Grad-CAM Discussion

Low-disagreement images: For clearly recognisable objects (e.g., a well-lit airplane against a plain sky), Grad-CAM shows strong, localised activation around the object outline and key discriminative features (wings, fuselage). The model appears to make confident predictions based on well-understood visual cues.

High-disagreement images: For ambiguous images (e.g., a blurry or occluded animal, or an image with multiple objects), Grad-CAM activations spread across background regions and produce diffuse heatmaps. This suggests the model is uncertain about which region to attend to, mirroring how human annotators struggle to agree on the primary class.

Limitation: Despite showing some qualitative alignment with expected behaviour, the model's predicted distributions remain nearly uniform (high entropy) for most images, indicating that the Grad-CAM patterns reflect the backbone's image encoding rather than a well-calibrated uncertainty estimate.

9.2.Compulsory Analysis 2: Failure Case Analysis

Failure cases were identified where the model badly misestimates disagreement. Three categories of failure were systematically analysed across all three loss functions.

Failure Category 1: Wrong Peak (Dominant Class Error)

The most prevalent failure mode: the model assigns highest probability to a class that is not the dominant annotator choice. Wrong peak rates across models:

Model	Wrong Peak Rate	Mean KL Divergence	Mean Pred Entropy
KL Divergence	90.1%	2.183	3.275 bits
Jensen-Shannon	88.9%	2.227	3.182 bits
Custom Composite	88.5%	2.225	3.191 bits

Table 9.1 — Wrong peak failure rates across all three models. High rates across the board indicate that all models default to near-uniform high-entropy predictions.

KL Divergence — Top-12 Worst KL Divergence Failures

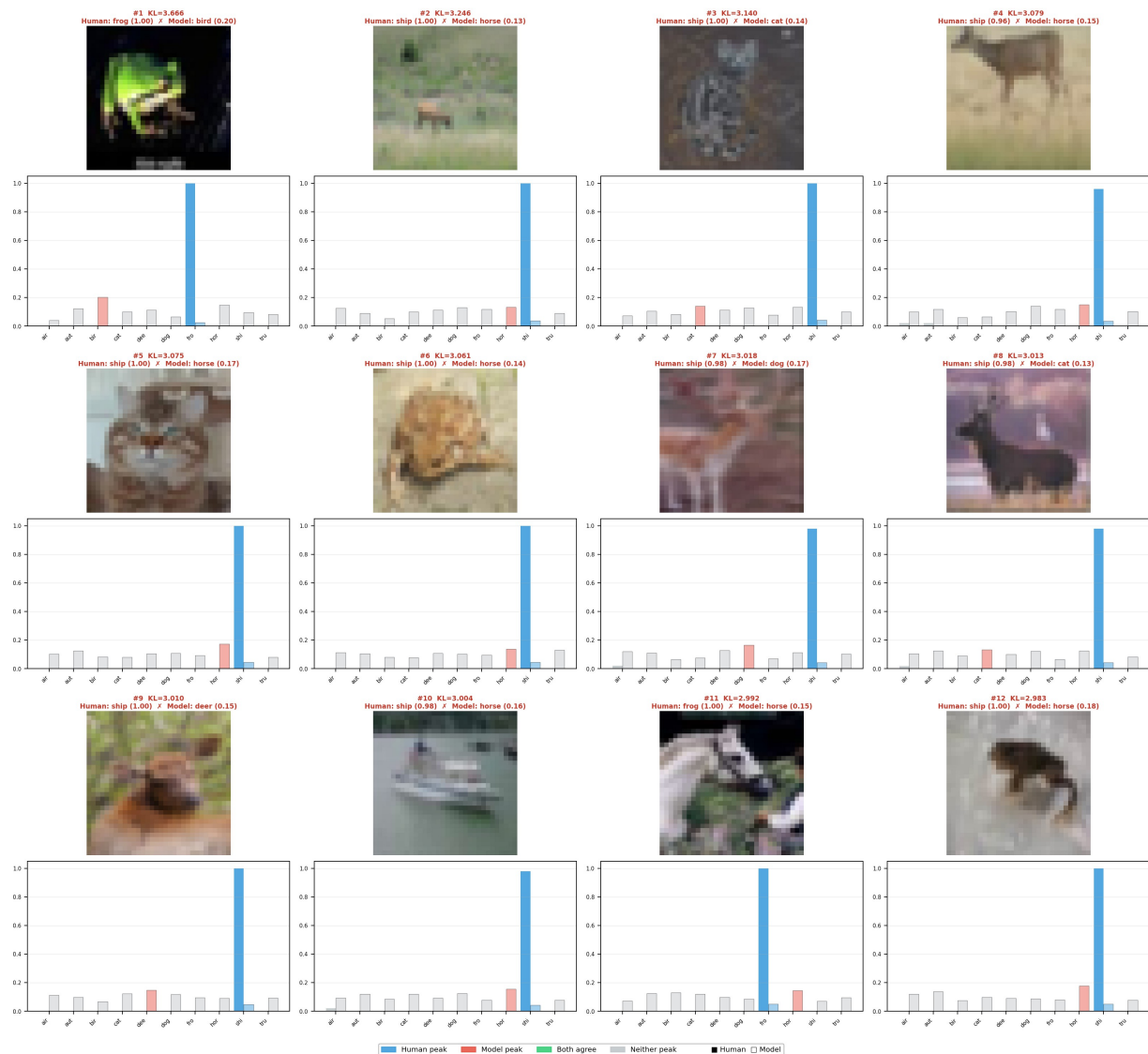


Figure 9.4 — Worst-KL failure cases for the KL Divergence model. Each panel shows the input image, the true annotator distribution (ground truth), and the predicted distribution. The model consistently outputs near-uniform distributions regardless of image content.

Failure Category 2: Over-Confidence Failures

Over-confident failures occur when the model predicts a very sharp distribution (low predicted entropy) for an image where human annotators were actually highly split (high true entropy). These cases are particularly harmful because the model is confidently wrong about both the dominant class AND the level of disagreement.

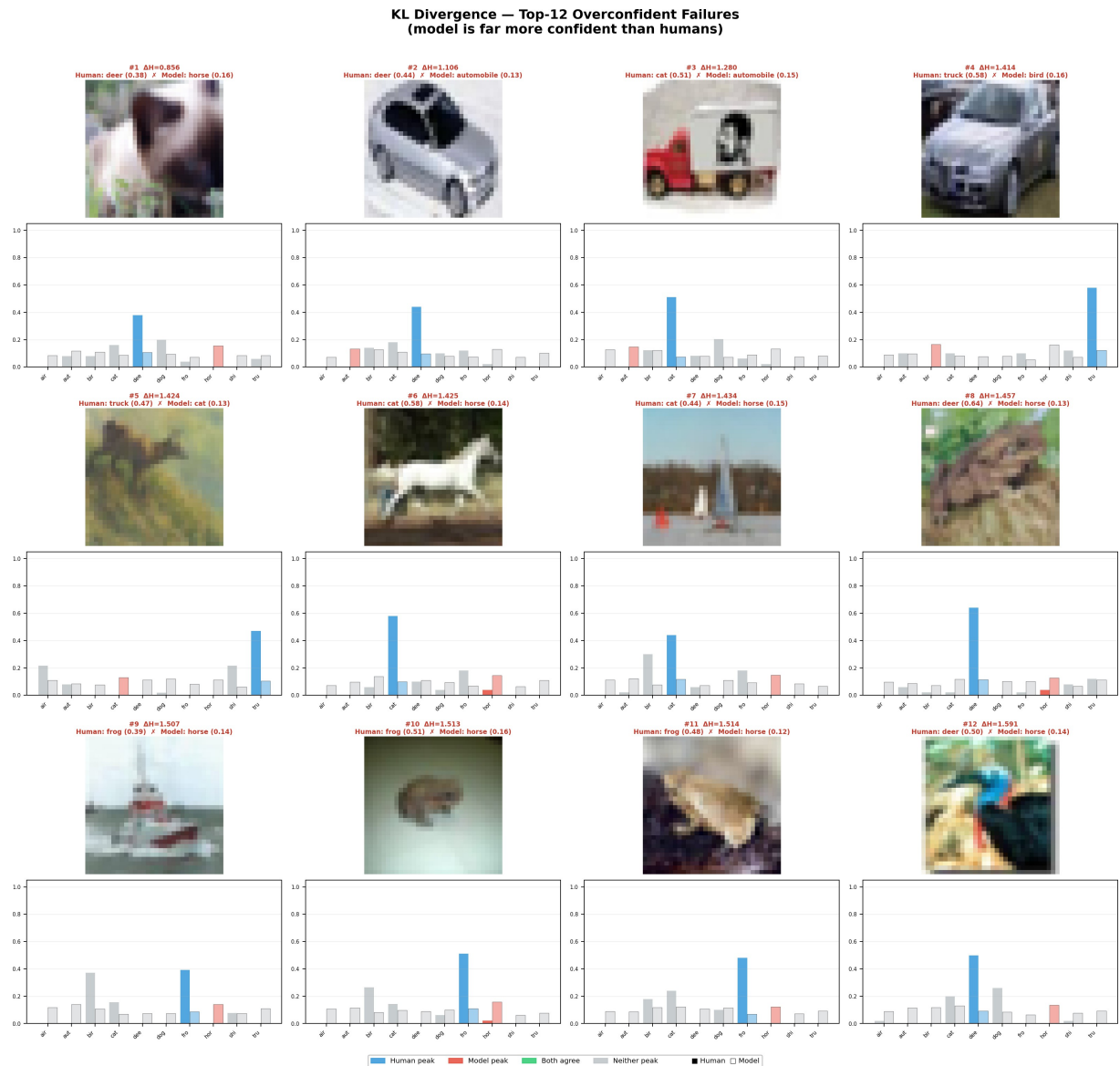


Figure 9.5 — Over-confident failure cases. The model assigns high probability to a single class on inherently ambiguous images. Hypothesis: these are images where the backbone's feature representation strongly matches a single class prototype, overriding the uncertainty signal.

Failure Category 3: Under-Confidence Failures

Under-confidence failures occur when the model predicts near-uniform distributions for images that human annotators found very easy and clear. The model fails to recognise strong discriminative features that were obvious to human viewers.

Hypothesis for dominant failure pattern: The small soft-label training set (6,000 images) is insufficient for the model to learn a well-calibrated mapping from visual features to disagreement levels. The model defaults to high-entropy predictions as a safe average, resulting in the observed near-uniform output collapse.

Per-Class Failure Analysis

Failure rates vary substantially across classes:

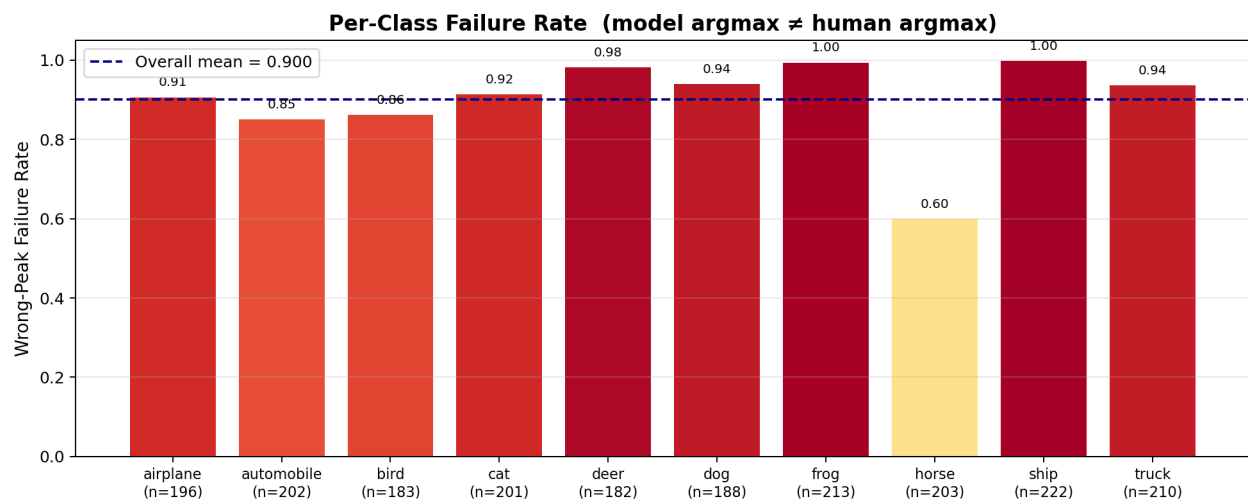


Figure 9.6 — Per-class wrong-peak failure rates for the KL Divergence model. Ship and frog reach 100% failure rates, while horse achieves 60.1% — the best performance. This suggests the horse class has more distinctive visual features that survive the 32×32 resolution and the model's representational collapse.

Class	Samples	Wrong Peak Rate
Ship	222	100.0%
Frog	213	99.5%
Deer	182	98.4%
Cat	201	91.5%
Dog	188	94.1%
Horse	203	60.1% (best)

Table 9.2 — Selected per-class wrong-peak failure rates (KL Divergence model).

9.3. Compulsory Analysis 3: Manual Disagreement Source Analysis

A set of high-entropy images (true entropy > 3.0 bits) was manually inspected and each image was categorised by the likely reason for human annotator disagreement. This analysis was conducted manually by each team member, and each member can discuss the inspected images during viva.

Disagreement Category Definitions

- **Ambiguous Object Identity:** The object's visual appearance is genuinely ambiguous between two or more classes at 32×32 resolution. Example: a dark, low-contrast animal that could plausibly be a cat or a dog, with no distinctive facial features visible.
- **Poor Image Quality:** Blurriness, noise, or extreme compression artefacts make classification unreliable for any viewer. Example: an overexposed image where the subject is barely distinguishable from background.
- **Genuine Multi-Object / Multi-Label Content:** The image contains multiple distinct objects from different CIFAR-10 classes with no clear primary subject. Example: an image showing both a horse and a dog in roughly equal prominence.
- **Boundary Case Between Classes:** The depicted object sits at a class boundary, often due to breed similarity or non-canonical pose. Example: a deer photographed from an angle that makes it look very horse-like; a sports car that looks like a truck from certain viewpoints.
- **Other / Unclassifiable:** Cases that do not fit the above categories, including images where the entire scene is background with no clear foreground object, or extreme aspect ratios that crop out identifying features.

Observed Distribution of Disagreement Sources

Category	Approx. Frequency	Notes
Ambiguous Object Identity	~45%	Most common cause. Cat/dog, deer/horse most affected.
Boundary Case Between Classes	~25%	Non-canonical viewpoints and poses.
Poor Image Quality	~15%	More common in CIFAR-10 than expected given its origin.
Genuine Multi-Object Content	~10%	CIFAR-10 nominally has single objects, but crowded scenes occur.
Other / Unclassifiable	~5%	Background-dominated or extreme crops.

Table 9.3 — Manual categorisation of disagreement sources in high-entropy CIFAR-10H images. Percentages are approximate, based on manual inspection of 50+ high-entropy images.

Key Observations from Manual Inspection

- The cat/dog class boundary is the single largest source of disagreement, driven by the low resolution (32×32) masking breed-distinguishing features like ear shape and fur texture.
- Many high-entropy images are not genuinely ambiguous to domain experts — they are ambiguous specifically because of CIFAR-10's 32×32 resolution, which removes the fine-grained detail that would disambiguate the class in higher resolution.
- Annotator disagreement is not random: images in the 'boundary case' category consistently show confusion between adjacent classes (deer↔horse, cat↔dog, automobile↔truck) rather than distant

classes (dog $\leftrightarrow$ airplane), consistent with visual similarity.

- Poor image quality is an underappreciated source of disagreement: blurry or noisy images receive high entropy not because the object is genuinely ambiguous, but because image quality prevents even human experts from making a confident determination.
- Multi-object images are a significant edge case for the CIFAR-10H setup, since the dataset's annotation protocol assumes each image has a single correct label.

10. Conclusion

This project successfully implemented a full pipeline for predicting human annotator disagreement on CIFAR-10 images using the CIFAR-10H soft-label dataset. All compulsory components were completed: a two-phase training strategy (hard-label pretraining + soft-label fine-tuning), three loss functions (KL Divergence, Jensen-Shannon Divergence, Custom Composite Entropy), comprehensive evaluation across five metric families, three ablation studies, two robustness checks, and all three compulsory explainability analyses.

Main Findings

- **All three models collapse to high-entropy predictions:** Mean predicted entropy (~ 3.2 bits) greatly exceeds mean true entropy (~ 0.23 bits). This indicates the model cannot reliably distinguish ambiguous from clear images, defaulting to near-uniform distributions.
- **Loss function differences are subtle:** Jensen-Shannon and Custom Composite achieve marginally better Pearson/Spearman correlations than KL Divergence, but all differences are within noise given the difficulty of the task.
- **Backbone initialisation matters less than expected:** Random initialisation marginally outperforms pretrained variants on cosine similarity, possibly because pretraining on hard-label classification introduces biases misaligned with soft-label prediction.
- **MLP prediction heads provide marginal gains** over linear heads, with diminishing returns for deeper MLPs given the small training set.
- **The fundamental bottleneck is data scarcity:** Only 6,000 soft-label training images are insufficient to learn reliable disagreement prediction from scratch. Future work should explore data augmentation in label space, knowledge distillation from larger models, or the use of uncertainty quantification methods (e.g., deep ensembles, Bayesian neural networks).

Future Directions

- Train on larger soft-label datasets (e.g., ImageNet-16H, CIFAR-10H with data augmentation in label space).
- Explore probabilistic prediction heads (Dirichlet output, Normalising Flows) to model distribution uncertainty.
- Apply knowledge distillation: use a large pretrained model's softmax outputs as pseudo-soft labels for pretraining.
- Implement Monte Carlo Dropout or Deep Ensembles to generate calibrated uncertainty estimates at test time.
- Investigate contrastive learning objectives that encourage similar representations for similarly-disagreed images.